Hybrid Quantum–Classical Neural Network for Pneumonia Detection in the NISQ Era

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1. Introduction & Scientific Context

Clinical motivation. Pneumonia causes ∼2.5 million deaths annually. Manual radiologist interpretation is subjective; deep learning (ResNet, DenseNet) achieves $> 85\%$ diagnostic accuracy but requires massive parameter counts (164.8k+ per decision layer).

Quantum computing opportunity. NISQ devices (6–100 qubits) offer parameter-efficient learning via Variational Quantum Algorithms (VQA). However, their effectiveness in medical imaging remains underexplored, especially hybrid architectures combining classical feature extraction with quantum classification [3, 8].

Research hypothesis: A quantum decision layer (VQC with ∼100 parameters) can approach classical discrimination while reducing memory footprint, enabling edge/IoT deployment.

2. Literature Review

Recent advances in QML and medical imaging:

- **Classical baseline:** ResNet for image recognition [2].
- **Quantum encoding:** encoding strategies and trade-offs [5].
- **VQC foundations:** quantum circuit learning [4].
- **Barren plateaus:** vanishing gradients in PQCs [7].
- **Dataset shift:** pneumonia generalization issues across sites [6].

Gap: Limited empirical studies quantifying the efficiency–accuracy trade-off of CNN→VQC hybrids under realistic distribution shifts.

3. Methodology: Hybrid Architecture

Design rationale. We separate *feature learning* (classical) from *decision making* (quantum) to stay within NISQ constraints.

Stage	Input	Operation	Output	Params
Feature extraction	224×224 image	ResNet-50 (frozen)	2048-D vector	0 (fixed)
Dimensionality reduction	2048-D	PCA (linear)	64-D vector	0 (fixed)
Quantum classification	64-D	VQC $U(\theta)$	score $[0, 1]$	108

Table 1: Table 1. Hybrid pipeline architecture and parameter counts.

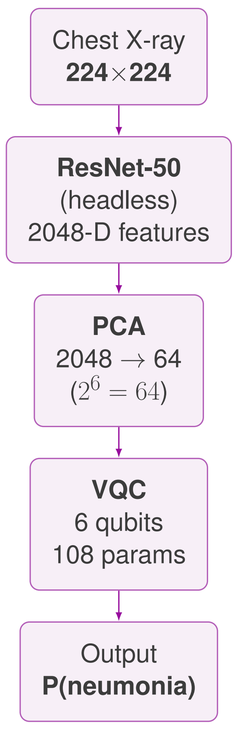

Figure 1: Figure 1. Implemented hybrid pipeline diagram (TikZ).

Feature extraction: ResNet-50 without classification head outputs a 2048-D feature vector.
PCA compression: 2048 → 64 dimensions to match 6-qubit amplitude encoding.
Quantum classifier: 6-qubit VQC with a shallow ansatz and 108 trainable parameters.

4. Quantum Circuit: Theory

Amplitude encoding: normalized vector $x \in \mathbb{R}^{64}$, $\|x\| = 1$, mapped to:

$$|\psi(x)\rangle = \sum_{i=0}^{63} x_i |i\rangle \in \mathcal{H}_{2^6} \tag{1}$$

VQC prediction: with ansatz $U(\theta)$ and observable Z_0:

$$f(x, \theta) = \langle \psi(x) | U^\dagger(\theta)\, Z_0\, U(\theta) | \psi(x) \rangle \in [-1, 1] \tag{2}$$

Probability mapping: $\sigma(f) = (f + 1)/2$.
Loss: binary cross-entropy, optimized via parameter-shift gradients [4].

5. Experimental Setup

Dataset: Guangzhou Chest X-ray dataset (624 images: 312 pneumonia, 312 normal).

- **Train/Test:** 70/30 split (437 train, 187 test).
- **Shift:** Train 74% pneumonia vs Test 62% pneumonia (robustness stress test).

Hyperparameters:

- Optimizer: Adam (lr = 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Epochs: 100, batch size: 16
- PCA dims: 64 (ablation: 8, 16, 32, 64, 128)
- Circuit depth: 4 layers

Simulation: IBM Qiskit (statevector simulator, noise-free).

Reproducibility: Code available at `github.com/michalforgo/qml-pneumonia`
Dataset: Guangzhou CXR dataset, preprocessing following [1]

6. Results: Parameter Efficiency

Parameter comparison:

Decision layer	# Parameters	Reduction
Classical MLP (1 hidden: 128)	164,865	1×
VQC (6 qubits, 4 layers)	108	**1527×**

Table 2: Table 2. Parameter reduction in the quantum decision head.

This reduction enables:

- Smaller memory footprint for the decision head.
- Faster optimization (fewer trainable parameters).
- Potential for edge deployment (subject to hardware feasibility).

7. Results: Diagnostic Performance

Test set (187 images, 62% pneumonia prevalence):

Metric	Hybrid (VQC)	Classical (MLP)
Sensitivity (recall)	80.00%	92.11%
Specificity	57.69%	76.92%
Accuracy	71.63%	85.90%
AUC	**0.74**	**0.88**

Table 3: Table 3. Diagnostic performance and efficiency–accuracy trade-off.

Observations:

- VQC reaches AUC 0.74 with 1527× fewer parameters in the head.
- Specificity drops under dataset shift + training difficulty (barren plateaus) [7, 6].

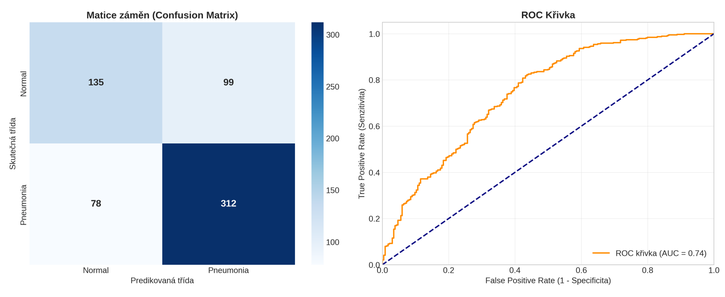

Figure 2: Figure 2. ROC curve and confusion matrix (your provided plot).

8. Ablation: PCA Dimensionality

Effect of PCA compression on performance:

PCA dims	Qubits	AUC	Params
8	3	0.62	27
16	4	0.68	48
32	5	0.71	75
64	**6**	**0.74**	**108**
128	7	0.73	147

Table 4: Table 4. PCA/qubit trade-off vs AUC.

Optimal here: 6 qubits (64 dims). Larger systems can become harder to train due to barren plateaus [7].

9. Limitations & Discussion

1) Barren plateaus. Gradients can vanish in PQCs; mitigation includes ansatz design and training strategies [7, 8].
2) Dataset shift. Prevalence differences can reduce specificity; mitigation includes calibration/domain adaptation [6].
3) Hardware noise. Real devices have decoherence and gate errors; mitigation includes error mitigation techniques.
4) PCA loss. Compression may discard useful variance; alternative encodings could be explored [5].

10. Conclusion & Future Work

This study demonstrates a functional hybrid quantum–classical pipeline for pneumonia detection. The VQC decision layer achieves **AUC 0.74 with 1527× parameter reduction**, supporting the case for parameter-efficient quantum decision heads in constrained settings.

Future directions:

- Run on real hardware with error mitigation.
- Evaluate on larger datasets (CheXpert / NIH).
- Improve robustness to dataset shift (calibration, balanced training).

References

References

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